Test Data & Data Masking

Ref: 4.6
Product: iConnect Plus
Organisation: Cache Digitech
Classification: Internal — Confidential
Version: 2.0
Date: 2026-11-03
Deployment model: On-premises (Cache Digitech premises) · VPN for remote access
Audience: QA, developers, platform engineering, DPO

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1. Purpose

Define rules for **non-production data**: synthetic fixtures, masked copies of production, and prohibitions against using live **restricted** personal data in dev without approval.

2. Environment data policy

Environment	Allowed data	VPN / network
Development	Synthetic seeds (`seed_demo_modules.py`), fabricated PII	Local / dev VLAN
Test / UAT	Masked subset or full synthetic	VPN for remote
Production	Live data	LAN / VPN only

Prod → Dev direct copy: **prohibited** unless explicit DPO + IT approval (2.3).

3. Masking requirements

When refreshing UAT from production backup, apply at minimum:

Field type	Masking rule
Email	Replace domain with `@example.invalid`; preserve uniqueness
Phone	Random 10-digit invalid range
PAN / Aadhaar / tax IDs	Replace with test values; re-encrypt with UAT Fernet keys
Bank account	Tokenize to non-routable test numbers
Names	Faker or fixed catalog — no real employee names
Passwords	Reset to known test password (bcrypt) — never prod hash
JWT / API secrets	Regenerate all secrets in UAT `.env`

Masking script / tool: Template / Pending sign-off (owner: Platform).

4. Synthetic data

Source	Use
Demo seed scripts	Local dev, demos
Factory builders in tests	CI unit/integration
Manual CSV imports	UAT scenarios — must pass masking checklist

5. Encrypted fields

Masked exports must either:

1. Decrypt with prod keys → mask → re-encrypt with UAT keys, or
2. Null out restricted columns in UAT refresh.

Never copy production `.env` encryption keys to UAT.

6. MinIO / attachments

Control	Action
Documents with PII	Exclude from UAT refresh or scrub binaries
Bucket	Separate UAT bucket/volume
ClamAV	Same as prod target in UAT

7. Access to test data

Role	Access
Developers	Dev synthetic only (default)
QA	UAT masked
Support	No standing prod DB access

8. Verification checklist (UAT refresh)

#	Check	Pass
1	No production secrets in UAT `.env`	
2	Sample of 20 users — emails masked	
3	No valid PAN/Aadhaar in clear text	
4	Audit log volume acceptable or truncated	
5	DPO sign-off on refresh ticket	

Last UAT refresh evidence: Template / Pending sign-off

9. Related documents

Ref	Document
2.3	Environment Separation
4.1	Data Inventory & Classification
4.5	Personal Data Processing Record

10. Change log

Date	Change	By
2026-11-03	v2.0 masking baseline	Platform